National University of Computer and Emerging Sciences, Lahore Campus				
	Course:	**Machine Learning for Robotics**	**Course Code:**	**CS 5096**
	Program:	**BS (Computer Science)**	**Semester:**	**Spring 2026**
	Date:	**04-05-2026**	**Total Marks:**	**15**
	Section:	**A**	**Weight:**	
	Exam:	**Quiz 3**	**Roll No:**	

Q1) Suppose we have 5 data points in a 2-dimensional space.

a) Let the initial cluster centroids be $\mu_1=(2,2)$ and $\mu_2=(8,7)$. **(5)**
Calculate the Euclidean distance from each data point to both centroids and assign each data point to the nearest cluster.

x_i	$d(x, \mu_1)$	$d(x, \mu_2)$	Cluster (C_1/C_2)
(1, 3)	1.414	8.06	C_1
(2, 5)	3	6.32	C_1
(4, 4)	2.83	5	C_1
(8, 2)	6	5	C_2
(9, 4)	7.28	3.16	C_2

1 mark each for each correct row

b) Compute the updated coordinates for the centroids. **(2)**

$\mu_1' = (\frac{7}{3}, 4)$

$\mu_2' = (8.5, 3)$

c) A new data point $x_6=(3,4)$ is added to the dataset and assigned to Cluster C_1. Calculate the coordinates of the updated centroid μ_1' including this new point. **(2)**

(2.5, 4)

Q2) The discounted return at time t is given by
$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \ldots.$
Suppose $\gamma = 0.5$ and the reward sequence is
$R_1 = 1$, $R_2 = 6$, $R_3 = 12$, $R_4 = 16$, then zero for R_5 and later.

a) Find G_3 and G_4. **(2)**

$G_3 = 16 + 0.5\ (0) = 16$

$G_4 = 0$

b) A robot is navigating a 4x4 grid to find a goal while avoiding obstacles. The goal is a specific square in the grid where the agent receives a positive reward.

i) Define the state space. **(2)**

The state space S represents all possible locations of the robot on the 4x4 grid. It can be represented by a set of coordinates (x,y), where x∈{0,1,2,3} and y∈{0,1,2,3}.
Other valid representation:

- Map every cell in the 4×4 grid to a single integer from 1 to 16.
  S∈{1,2,…,16}

ii) Is the action space discrete or continuous? Justify your answer. **(2)**

The action space is discrete because the robot chooses from a finite, distinct set of movements (e.g., Up, Down, Left, Right) rather than choosing from an infinite range of continuous values (like a specific angle or speed).
1 mark for correct action space
1 mark for justification